Mohamed Gamil

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SUMMARY

Junior .NET Backend Engineer experienced in building scalable RESTful APIs with C#, ASP.NET Core, EF Core, and SQL Server. Skilled in Clean Architecture, Redis caching, RabbitMQ, Hangfire, and AI-assisted backend integrations.

PROJECTS

Incident Intelligence Platform–AI-powered incident management backend *GitHub* 08/2026 – Present

- Developed a scalable Incident Intelligence Platform backend utilizing **ASP.NET Core**, Entity Framework Core, and SQL Server, featuring **JWT authentication**, **role-based access control**, and structured Serilog logging
- Enhanced system performance and modularity by implementing **Redis** cache-aside caching, **Hangfire** background processing, and a **RabbitMQ** event-driven architecture
- Integrated an automated **AI root-cause analysis** pipeline utilizing LLM APIs and Retrieval-Augmented Generation (**RAG**) to generate diagnostic evidence and recommended remediation actions for system incidents
- Modeled complex business domains and operational workflows by implementing custom incident **state machines** for full timeline tracking and automated log correlation rules.

Faculty Operating System (FOS)– Kernel Memory & Process Management *GitHub* 10/2026 – 12/2026

- Implemented kernel and user-level heap allocators (**kmalloc/kfree**) featuring **exact-fit and worst-fit** allocation strategies capable of handling 1000+ allocations under stress tests
- Designed efficient memory segment management utilizing linked lists and **segment merging** mechanisms to minimize **external fragmentation**
- Optimized memory concurrency and synchronization using **spinlocks and semaphores**

ServiMarket–Flutter Mobile App *GitHub* 3/2026 – 7/2026

- Developed a scalable service marketplace app using **Flutter** and **Firebase** as a **DEPI final project**, implementing **GoRouter** and **BLoC** state management
- Engineered real-time features including instant messaging via Firestore streams, GPS location-aware sorting, and keyword-based ad filtering
- Integrated third-party cloud storage (Cloudinary) for media uploads and automated user profile handling

SKILLS

Clean Architecture & RESTful APIs | Event-Driven Architecture (RabbitMQ) |
Distributed Systems Concepts (Idempotency, Retries, Dead Letter Queues) |
Caching Strategies (Redis Cache-Aside, TTL) | Background Processing (Hangfire) |
Database Management & ORMs (Entity Framework Core, SQL Server, MongoDB) |
Firebase (Auth, Firestore, Cloud Messaging) | Docker & Containerization | Git & GitHub

EDUCATION

Faculty of Computer and Information Sciences, *Ain Shams University* 07/2023 – Present | Egypt,Giza

- **Relevant Coursework**: Operating Systems, Data Structures & Algorithms, System Design, Database Systems, Computer Networks, Software Engineering.

COURSES

Backend Developer Trainee, *Information Technology Institute (ITI)* 08/2026 – 09/2026 | Giza , Egypt

- Intensive training in .NET ecosystem, object-oriented C#, and relational database design

Cross-Platform Mobile Development Trainee, *Digital Egypt Pioneers Initiative (MCIT* 11/2025 – 07/2026 | Giza , Egypt

- Ministry of Communications and Information Technology)

- Intensive scholarship focused on cross-platform engineering, Firebase integrations, and state management